Samuele Punzo

AI Master's student, University of Amsterdam
Researcher in Interpretability & xAI

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Google scholar Profile

Research Interests

Mechanistic Interpretability, LLMs and VLMs explainability, AI safety

Research Experience

Supervised Program for Alignment Research (SPAR) *Sep. 2026 – present*
Fellow Researcher remote

– Working on locating the refusal circuit in encoder-free models under the supervision of Alessandro Suglia (University of Edinburgh) and Francesco Pinto (Google DeepMind).

Max-Planck Institute for Informatics (MPI-INF) *July 2026 – present*
Visiting Researcher in the Explainable Machine Learning group (Prof. Jonas Fischer) Saarbrücken, DE

– Exploring interpretability of VLMs applied to 3D scenes using sparse autoencoders for concept extraction. This work led to the discovery of 3D spatially-grounded explicit representations that can make downstream tasks like VQA, and view-selection inherently interpretable.

Institute for Logic, Language and Computation (ILLC) — UvA *Jan. 2026 – July 2026*
Research Assistant in the MULINI Lab (Prof. Sandro Pezzelle) Amsterdam, NL

– Discovered circuits for Animacy completion across 4 LLM architectures (Llama, Qwen, Gemma, GPT-2) using **TransformerLens** and tested their transferability across similar formulations of the task.
– Proposed a novel token-level perplexity framework to test whether LLMs rely on expected linguistic cues when solving tasks.
– **Open source contributions:** *TransformerLens*, *EAP-ig*

BioMedical and Health Informatics Lab — University of Pisa *Nov. 2024 – Nov. 2025*
Research Assistant in the BioMedical and Health Informatics Lab (Prof. Alina Sirbu) Pisa, IT

– **MEDICA and PANS projects**:
  ∗ Developed explainable machine learning pipelines for Microarray and scRNA-seq data to identify novel Multiple Sclerosis and PANS biomarkers and pathways, using interpretability frameworks and biological enrichment analysis tools.

Education

University of Amsterdam *Sep. 2025 – July 2027 (expected)*
M.S. in Artificial Intelligence | Link to all courses GPA: 8.8/10 (**Excellent**, A+ in US/UK scale)

University of Pisa *Sep. 2021 – Oct. 2024*
B.S. in Computer Science | Link to all courses Grade: 110/110 with honours | GPA: 29/30

Bachelor Thesis: *Research of new biomarkers for Multiple Sclerosis using Machine Learning techniques*

Honours and Awards

Lead The Future
Selected as a mentee (acceptance rate <15%) for a leading STEM mentorship non-profit organisation.
Top 5% of Students
Among the 5% best students in all three academic years of the BSc at University of Pisa.

Publications

Do Video Foundation Models Understand Intuitive Physics? A layer-wise probing analysis *2026*
S. Punzo*, N. Caselli*, S. Lo Sardo, I. Pantelidis, F. Massafra (*equal contribution)* arXiv:2606.09646
NeurIPS 2026 Workshop on World Models in Physical AI

Where Animacy Lives in Large Language Models: Tracing the Circuits of the Animacy Concept *2026*
S. Punzo*, G. Cinà, S. Pezzelle* arXiv:2607.20995
Accepted at the 9th BlackboxNLP Workshop @ EMNLP'26

Is my model perplexed for the right reason? Contrasting LLMs' Benchmark Behavior with Token-Level Perplexity *2026*
Z. Prins, **S. Punzo***, F. Wildenburg*, G. Cinà, S. Pezzelle (*equal contribution)* arXiv:2603.29396
EMNLP'26 - Conference on Empirical Methods in Natural Language Processing

[Re] Towards Safer Pretraining: Analyzing and Filtering Harmful Content in Web-Scale Datasets for Responsible LLMs *Under review*
N. Caselli, S. Lo Sardo*, F. Massafra*, **S. Punzo*** *(*equal contribution)*

Mind the Gap: Promise and Pitfalls of Hierarchical Planning in LeWorldModel *2026*
N. Caselli, S. **S. Punzo***, F. Massafra*, S. Lo Sardo, I. Pantelidis (*equal contribution)* arXiv:2607.12547
NeurIPS 2026 PTA: From Pretrained Representations to Acting Agents Workshop

Machine Learning for analysis of Multiple Sclerosis cross-tissue bulk and single-cell transcriptomics data *Under review*
F. Massafra, **S. Punzo***, S.G. Galfrè, A. Maglione, C. Priami, A. Sîrbu (*equal contribution)* arXiv:2603.05572

ML Pipeline for Multiple Sclerosis Biomarker Discovery: XAI vs. Statistical Approaches *2025*
S. Punzo, S.G. Galfrè, F. Massafra, A. Maglione, C. Priami, A. Sîrbu* arXiv:2509.22484

– Presented at BBCC24, CIBB25.

Teaching Experience

Vrije Universiteit Amsterdam *Feb. - Mar. 2026*
Teaching Assistant — BSc Machine Learning course Amsterdam, NL

– Delivered two exercise lectures and one project session per week, guiding 10 groups of students through the development of their machine learning projects.

Extracurricular Activities

ARENA course on AI Safety (*Self-study*) *Jun. 2026 – Aug. 2026*

Bologna Orthogonal Summer Term '24 (*Summer School*) *Aug. 2024*

Futuro Annunciato '24 (*Summer School*) *Jul. 2024*

Languages

Italian: Native, **English**: C1 Cambridge Certificate, **German**: A2